Sahaj Bhadja

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Summary

Software engineer with 2+ years building **React/TypeScript interfaces**, **Python APIs**, and **AI-powered products**, with hands-on experience shipping scalable full-stack features, cloud-backed workflows, and user-facing systems in fast-moving environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL access patterns, and improving responsiveness for analytics workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving runtime observability, release confidence, and robustness across recurring production data-processing workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable backend services and responsive interfaces while improving delivery predictability through clearer ownership and sequencing.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production-scale operations with fault-aware batch execution and CI/CD-backed release practices.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation and high-confidence benchmarking while improving signal quality for debugging model-driven decision workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for high-volume user-facing processes.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation, troubleshooting, and handoffs.

Skills

Languages: **TypeScript**, **Python**, JavaScript, Go-adjacent backend exposure, SQL, Java, C++, Bash
Frontend: React, Next.js, component-driven interfaces, responsive web workflows, UX-focused implementation
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, ETL pipelines, distributed workflows, cloud-backed services
Data & Cloud: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Docker, Kubernetes, Terraform basics
Engineering: CI/CD, Jenkins, automated testing, technical documentation, Agile/Scrum, cross-functional collaboration

Projects

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an **AI-powered Next.js SaaS platform** for image editing and search, integrating Stripe billing and MongoDB-backed workflows while improving premium conversion by **18%** through polished experiences.

CaptionCraft Live | *Python, Transformers, Flask*
- Built an **AI-driven image captioning workflow** using transformer-based models, improving caption accuracy by **15%** while supporting scalable inference behavior through Flask-backed experimentation and evaluation tooling.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production ETL workflows.